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%% Questão 1 - Sinais Senoidais
% Intervalo de tempo (0 a 1 segundo, passo 0.001)
t = 0:0.001:1;

% (a) A=5, w=2π, φ=0
A = 5; w = 2*pi; phi = 0;
x1 = A * sin(w*t + phi);

% (b) A=10, w=20π, φ=π/2
A = 10; w = 20*pi; phi = pi/2;
x2 = A * sin(w*t + phi);

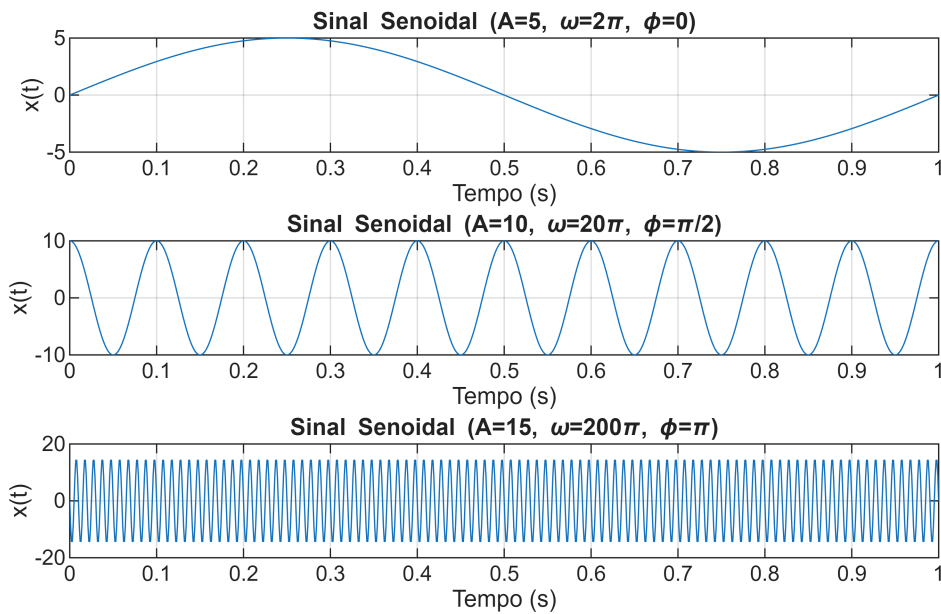
% (c) A=15, w=200π, φ=π
A = 15; w = 200*pi; phi = pi;
x3 = A * sin(w*t + phi);

% Plotando
figure;
subplot(3,1,1);
plot(t, x1); grid on;
title('Sinal Senoidal (A=5, \omega=2\pi, \phi=0)');
xlabel('Tempo (s)'); ylabel('x(t)');

subplot(3,1,2);
plot(t, x2); grid on;
title('Sinal Senoidal (A=10, \omega=20\pi, \phi=\pi/2)');
xlabel('Tempo (s)'); ylabel('x(t)');

subplot(3,1,3);
plot(t, x3); grid on;
title('Sinal Senoidal (A=15, \omega=200\pi, \phi=\pi)');
xlabel('Tempo (s)'); ylabel('x(t)');

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%% Questão 2 - Sinais Exponenciais
% Intervalo de tempo (0 a 10 segundos, passo 0.01)
t = 0:0.01:10;
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% (a)  $x(t) = e^{5t}$ 
x4 = exp(5*t);
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% (b)  $x(t) = e^{-5t}$ 
x5 = exp(-5*t);
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% (c)  $x(t) = e^{-t} \cos(10\pi t)$ 
x6 = exp(-t) .* cos(10*pi*t);
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% Plotando
figure;
subplot(3,1,1);
plot(t, x4); grid on;
title('Sinal  $x(t) = e^{5t}$ ');
xlabel('Tempo (s)'); ylabel('x(t)');
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```
subplot(3,1,2);
plot(t, x5); grid on;
title('Sinal  $x(t) = e^{-5t}$ ');
xlabel('Tempo (s)'); ylabel('x(t)');
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```
subplot(3,1,3);
plot(t, x6); grid on;
title('Sinal  $x(t) = e^{-t} \cos(10\pi t)$ ');
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xlabel('Tempo (s)'); ylabel('x(t)');
```

